

Requirements Traceability Matrix

No	Requirement Description	Module	Backend API	Frontend Page	LINQ Usage	Test Case ID	Status
FR-01	User shall be able to register with email verification OTP	Auth	POST /api/auth/register, POST /api/auth/verify-email	Register Page	No		Implemented
FR-02	User shall be able to login with JWT authentication	Auth	POST /api/auth/login	Login Page	No		Implemented
FR-03	User shall be able to request password reset with OTP verification	Auth	POST /api/auth/forgot-password, POST /api/auth/verify-reset-token, POST /api/auth/reset-password-with-token	Forgot Password Page	No		Implemented
FR-04	Admin shall be able to manage users (CRUD, approve registrations)	User Management	GET/POST/PUT/DELETE /api/users/*	User Management Page	Yes - Filtering		Implemented
FR-05	Customer shall be able to add and manage vehicles	Vehicle Management	GET/POST/PUT/DELETE /api/vehicles/*	Vehicle Management Page	Yes - Filtering		Implemented
FR-06	ServiceManager shall be able to view all vehicles	Vehicle Management	GET /api/vehicles	Vehicle List Page	Yes - Pagination		Implemented
FR-07	ServiceManager shall be able to manage service categories	Service Categories	GET/POST/PUT/DELETE /api/service-categories/*	Service Categories Page	Yes - Filtering		Implemented
FR-08	Customer shall be able to book a service request	Service Request	POST /api/service-requests	Book Service Page	No		Implemented
FR-09	Customer shall be able to view their service request history	Service Request	GET /api/service-requests/my-requests	My Requests Page	Yes - Filtering		Implemented
FR-10	ServiceManager shall be able to assign technicians to service requests	Service Assignment	POST /api/service-assignments	Service Assignment Page	Yes - Workload Query		Implemented
FR-11	Technician shall be able to view assigned tasks	Service Task	GET /api/service-tasks/my-tasks	Technician Dashboard	Yes - GroupBy Status		Implemented
FR-12	Technician shall be able to update task status and completion	Service Task	PUT /api/service-tasks/{id}	Task Update Page	No		Implemented
FR-13	ServiceManager shall be able to manage parts inventory	Parts Management	GET/POST/PUT/DELETE /api/parts/*	Parts Inventory Page	Yes - Stock Filtering		Implemented
FR-14	ServiceManager shall be able to create and manage part orders	Part Orders	GET/POST/PUT/DELETE /api/part-orders/*	Part Orders Page	Yes - Status GroupBy		Implemented

FR-15	ServiceManager/Technician shall be able to add parts to service requests	Service Parts	POST /api/service-parts	Service Parts Page	No		Implemented
FR-16	ServiceManager shall be able to generate bills for completed services	Billing	POST /api/bills/generate/{serviceRequestId}	Billing Page	Yes - Cost Aggregation		Implemented
FR-17	Customer shall be able to view bills and make payments	Payments	GET /api/bills/my-bills, POST /api/payments	Payment Page	No		Implemented
FR-18	ServiceManager shall be able to record payments	Payments	POST /api/payments	Payment Recording Page	No		Implemented
FR-19	System shall display dashboard with service statistics	Dashboard	GET /api/dashboard/*	Dashboard Page	Yes - Multiple GroupBy		Implemented
FR-20	System shall generate comprehensive reports	Reports	GET /api/reports/*	Reports Page	Yes - Aggregation		Implemented

Screen-wise Field Details

1. Login Screen

Field Name	Data Type	Mandatory	Validation Rules	Model/DTO
Email	String	Yes	Valid email format ([EmailAddress])	LoginDto.Email
Password	String	Yes	Required, non-empty	LoginDto.Password

2. Register Screen

Field Name	Data Type	Mandatory	Validation Rules	Model/DTO
First Name	String	Yes	Required, Max 50 characters	RegisterDto.FirstName
Last Name	String	Yes	Required, Max 50 characters	RegisterDto.LastName
Email	String	Yes	Valid email format ([EmailAddress]), Max 100 chars	RegisterDto.Email
Password	String	Yes	Min 6 characters	RegisterDto.Password

Phone Number	String	Yes	Valid phone format ([Phone]), Max 20 chars	RegisterDto.PhoneNumber
Role	String	Yes	Enum: Customer, Technician, ServiceManager	RegisterDto.Role

3. Forgot Password Screen

Field Name	Data Type	Mandatory	Validation Rules	Model/DTO
Email	String	Yes	Valid email format ([EmailAddress])	ForgotPasswordDto.Email

4. Reset Password Screen

Field Name	Data Type	Mandatory	Validation Rules	Model/DTO
Email	String	Yes	Valid email format	ResetPasswordWithTokenDto.Email
OTP Token	String	Yes	6-digit token	ResetPasswordWithTokenDto.Token
New Password	String	Yes	Min 6 characters	ResetPasswordWithTokenDto.NewPassword
Confirm Password	String	Yes	Must match New Password ([Compare])	ResetPasswordWithTokenDto.ConfirmPassword

5. User Management Screen

Field Name	Data Type	Mandatory	Validation Rules	Model/DTO
User ID	Integer	Auto	Primary Key, Auto-generated	UserDto.UserId
First Name	String	Yes	Required, Max 50 characters	CreateUserDto.FirstName
Last Name	String	Yes	Required, Max 50 characters	CreateUserDto.LastName
Email	String	Yes	Valid email format, Max 100 chars, Unique	CreateUserDto.Email
Password	String	Yes	Min 6 characters (Create only)	CreateUserDto.Password
Phone Number	String	Yes	Valid phone format, Max 20 chars	CreateUserDto.PhoneNumber
Role	String	Yes	Enum: Admin, ServiceManager, Technician, Customer	CreateUserDto.Role
Is Active	Boolean	No	Default: true	UpdateUserDto.IsActive
Is Pending Approval	Boolean	Auto	Default: true for Technician/ServiceManager	UserDto.IsPendingApproval
Created At	DateTime	Auto	Auto-generated on creation	UserDto.CreatedAt

6. Vehicle Management Screen

Field Name	Data Type	Mandatory	Validation Rules	Model/DTO
Vehicle ID	Integer	Auto	Primary Key, Auto-generated	VehicleDto.VehicleId

Registration Number	String	Yes	Required, Max 20 characters, Unique per customer	CreateVehicleDto.RegistrationNumber
Make	String	Yes	Required, Max 50 characters	CreateVehicleDto.Make
Model	String	Yes	Required, Max 50 characters	CreateVehicleDto.Model
Year	Integer	Yes	Range: 1900-2100	CreateVehicleDto.Year
Vehicle Type	String	No	Max 30 characters (e.g., Sedan, SUV, Truck)	CreateVehicleDto.VehicleType
Color	String	No	Max 30 characters	CreateVehicleDto.Color
Fuel Type	String	No	Max 20 characters (e.g., Petrol, Diesel, Electric)	CreateVehicleDto.FuelType
RC Number	String	No	Max 50 characters (Registration Certificate)	CreateVehicleDto.RcNumber
Owner ID	Integer	Auto	Foreign Key to Users (Customer)	VehicleDto.OwnerId
Owner Name	String	Auto	Computed from User entity	VehicleDto.OwnerName
Is Active	Boolean	No	Default: true	VehicleDto.IsActive

Created At	DateTime	Auto	Auto-generated on creation	VehicleDto.CreatedAt
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7. Service Category Screen

Field Name	Data Type	Mandatory	Validation Rules	Model/DTO
Category ID	Integer	Auto	Primary Key, Auto-generated	ServiceCategoryDto.CategoryId
Category Name	String	Yes	Required, Max 100 characters, Unique	CreateServiceCategoryDto.CategoryName
Description	String	No	Max 500 characters	CreateServiceCategoryDto.Description
Base Price	Decimal	Yes	Range: 0 to Max ([Range(0, ...)])	CreateServiceCategoryDto.BasePrice
Estimated Duration (Minutes)	Integer	Yes	Range: 1-10080 (1 min to 7 days)	CreateServiceCategoryDto.EstimatedDuration Minutes
Is Active	Boolean	No	Default: true	ServiceCategoryDto.IsActive

8. Book Service Screen (Service Request)

Field Name	Data Type	Mandatory	Validation Rules	Model/DTO
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Service Request ID	Integer	Auto	Primary Key, Auto-generated	ServiceRequestDto.ServiceRequestId
Vehicle ID	Integer	Yes	Required, Must belong to customer	CreateServiceRequestDto.VehicleId
Category IDs	Integer[]	Yes	Required, At least one category	CreateServiceRequestDto.CategoryIds
Issue Description	String	Yes	Required, Max 1000 characters	CreateServiceRequestDto.IssueDescription
Priority	String	Yes	Enum: Normal, Urgent	CreateServiceRequestDto.Priority
Scheduled Date	DateTime	No	Optional, Must be future date	CreateServiceRequestDto.ScheduledDate
Customer Remarks	String	No	Max 500 characters	CreateServiceRequestDto.CustomerRemarks
Status	String	Auto	Enum: Requested, Assigned, InProgress, Completed, Closed, Cancelled	ServiceRequestDto.Status

Requested Date	DateTime	Auto	Auto-generated on creation	ServiceRequestDto.RequestedDate
Completed Date	DateTime	Auto	Set when status = Completed	ServiceRequestDto.CompletedDate
Estimated Cost	Decimal	Auto	Sum of category base prices	ServiceRequestDto.EstimatedCost
Actual Cost	Decimal	Auto	Calculated from bill	ServiceRequestDto.ActualCost

9. Service Assignment Screen

Field Name	Data Type	Mandatory	Validation Rules	Model/DTO
Assignment ID	Integer	Auto	Primary Key, Auto-generated	ServiceAssignmentDto.AssignmentId
Service Request ID	Integer	Yes	Required, Valid ServiceRequest	CreateServiceAssignmentDto.ServiceRequestId
Technician ID	Integer	Yes	Required, Must be active Technician	CreateServiceAssignmentDto.TechnicianId
Assigned By ID	Integer	Auto	From authenticated ServiceManager	ServiceAssignmentDto.AssignedByUserId
Notes	String	No	Max 500 characters	CreateServiceAssignmentDto.Notes

Status	String	Auto	Enum: Assigned, InProgress, Completed	ServiceAssignmentDto.Status
Assigned Date	DateTime	Auto	Auto-generated on creation	ServiceAssignmentDto.AssignedDate
Started Date	DateTime	Auto	Set when technician starts work	ServiceAssignmentDto.StartedDate
Completed Date	DateTime	Auto	Set when status = Completed	ServiceAssignmentDto.CompletedDate

10. Technician Task Screen (Service Tasks)

Field Name	Data Type	Mandatory	Validation Rules	Model/DTO
Service Task ID	Integer	Auto	Primary Key, Auto-generated	ServiceTaskDto.ServiceTaskId
Service Request ID	Integer	Auto	Foreign Key to ServiceRequests	ServiceTaskDto.ServiceRequestId
Description	String	Auto	Task description (from category)	ServiceTaskDto.Description
Is Completed	Boolean	No	Default: false	UpdateServiceTaskDto.IsCompleted
Completed Date	DateTime	Auto	Set when IsCompleted = true	ServiceTaskDto.CompletedDate

Completed By User ID	Integer	Auto	Technician who completed the task	ServiceTaskDto.CompletedByUserId
Price	Decimal	No	Range: 0 to Max	UpdateServiceTaskDto.Price
Order Index	Integer	Auto	Order of task execution	ServiceTaskDto.OrderIndex

11. Parts Inventory Screen

Field Name	Data Type	Mandatory	Validation Rules	Model/DTO
Part ID	Integer	Auto	Primary Key, Auto-generated	PartDto.PartId
Part Name	String	Yes	Required, Max 100 characters	CreatePartDto.PartName
Part Number	String	Yes	Required, Max 50 characters, Unique	CreatePartDto.PartNumber
Description	String	No	Max 500 characters	CreatePartDto.Description
Unit Price	Decimal	Yes	Range: 0.01 to Max	CreatePartDto.UnitPrice
Quantity In Stock	Integer	Yes	Range: 0 to Max	CreatePartDto.QuantityInStock
Reorder Level	Integer	Yes	Range: 0 to Max	CreatePartDto.ReorderLevel
Supplier	String	No	Max 100 characters	CreatePartDto.Supplier
Is Active	Boolean	No	Default: true	PartDto.IsActive

Created At	DateTime	Auto	Auto-generated on creation	PartDto.CreatedAt
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12. Part Orders Screen

Field Name	Data Type	Mandatory	Validation Rules	Model/DTO
Part Order ID	Integer	Auto	Primary Key, Auto-generated	PartOrderDto.PartOrderId
Order Number	String	Auto	Auto-generated (PO-YYYYMMDD-XXX)	PartOrderDto.OrderNumber
Part ID	Integer	Yes	Required, Valid Part	CreatePartOrderDto.PartId
Quantity	Integer	Yes	Range: 1 to Max ([Range(1, ...)])	CreatePartOrderDto.Quantity
Unit Price	Decimal	No	Range: 0 to Max (defaults to part price)	CreatePartOrderDto.UnitPrice
Total Amount	Decimal	Auto	Calculated: Quantity × Unit Price	PartOrderDto.TotalAmount
Status	String	Auto	Enum: Pending, Ordered, Shipped, Delivered, Cancelled	PartOrderDto.Status

Order Date	DateTime	Auto	Auto-generated on creation	PartOrderDto.OrderDate
Expected Delivery Date	DateTime	No	Optional future date	CreatePartOrderDto.ExpectedDeliveryDate
Delivered Date	DateTime	Auto	Set when status = Delivered	PartOrderDto.DeliveredDate
Ordered By User ID	Integer	Auto	From authenticated ServiceManager	PartOrderDto.OrderedByUserId
Notes	String	No	Max 500 characters	CreatePartOrderDto.Notes

13. Service Parts Screen

Field Name	Data Type	Mandatory	Validation Rules	Model/DTO
Service Part ID	Integer	Auto	Primary Key, Auto-generated	ServicePartDto.ServicePartId
Service Request ID	Integer	Yes	Required, Valid ServiceRequest	AddServicePartDto.ServiceRequestId
Part ID	Integer	Yes	Required, Valid Part with stock	AddServicePartDto.PartId
Quantity	Integer	Yes	Range: 1 to available stock	AddServicePartDto.Quantity

Unit Price	Decimal	Auto	From Part entity at time of addition	ServicePartDto.UnitPrice
Total Price	Decimal	Auto	Calculated: Quantity × Unit Price	ServicePartDto.TotalPrice
Added By User ID	Integer	Auto	From authenticated user	ServicePartDto.AddedByUserId
Added At	DateTime	Auto	Auto-generated on creation	ServicePartDto.AddedAt

14. Billing Screen

Field Name	Data Type	Mandator y	Validation Rules	Model/DTO
Bill ID	Integer	Auto	Primary Key, Auto-generated	BillDto.BillId
Bill Number	String	Auto	Auto-generated (BILL-YYYYMMDD-XXX)	BillDto.BillNumber
Service Request ID	Integer	Auto	Foreign Key to ServiceRequests	BillDto.ServiceRequestId
Service Charge	Decimal	Auto	Sum of task prices (LINQ aggregation)	BillDto.ServiceCharge
Parts Charge	Decimal	Auto	Sum of service parts (LINQ aggregation)	BillDto.PartsCharge
Subtotal	Decimal	Auto	ServiceCharge + PartsCharge	BillDto.Subtotal

Tax Percentage	Decimal	No	Range: 0-100	GenerateBillDto.TaxPercentage
Tax	Decimal	Auto	Subtotal × (TaxPercentage / 100)	BillDto.Tax
Discount	Decimal	No	Range: 0 to Subtotal	GenerateBillDto.Discount
Total Amount	Decimal	Auto	Subtotal + Tax - Discount	BillDto.TotalAmount
Amount Paid	Decimal	Auto	Sum of payments (LINQ aggregation)	BillDto.AmountPaid
Balance Due	Decimal	Auto	TotalAmount - AmountPaid	BillDto.BalanceDue
Payment Status	String	Auto	Enum: Unpaid, PartiallyPaid, Paid	BillDto.PaymentStatus
Due Date	DateTime	Yes	Required, Must be future date	GenerateBillDto.DueDate
Generated Date	DateTime	Auto	Auto-generated on creation	BillDto.GeneratedDate
Generated By User ID	Integer	Auto	From authenticated ServiceManager	BillDto.GeneratedByUserId

15. Payment Screen

Field Name	Data Type	Mandatory	Validation Rules	Model/DTO
Payment ID	Integer	Auto	Primary Key, Auto-generated	PaymentDto.PaymentId

Bill ID	Integer	Yes	Required, Valid Bill with balance	CreatePaymentDto.BillId
Amount	Decimal	Yes	Range: 0.01 to balance due	CreatePaymentDto.Amount
Payment Method	String	Yes	Required (e.g., Cash, Card, UPI, NetBanking)	CreatePaymentDto.PaymentMet hod
Transactio n ID	String	No	Optional, For online payments	CreatePaymentDto.TransactionId
Payment Status	String	Auto	Enum: Pending, Paid, Failed, Refunded	PaymentDto.PaymentStatus
Payment Date	DateTim e	Auto	Auto- generated on creation	PaymentDto.PaymentDate
Recorded By User ID	Integer	Auto	From authenticate d user	PaymentDto.RecordedByUserId
Notes	String	No	Optional payment notes	PaymentDto.Notes

16. Dashboard Screen

Field Name	Data Type	Mandato ry	Validati on Rules	Model/DTO
Total Services	Integer	Auto	LINQ Count()	DashboardReportDto.ServiceStatusSummary.T otalActive

Pending Services	Integer	Auto	LINQ Where + Count	ServiceStatusSummaryDto.Pending
In Progress Services	Integer	Auto	LINQ Where + Count	ServiceStatusSummaryDto.InProgress
Completed Services	Integer	Auto	LINQ Where + Count	ServiceStatusSummaryDto.Completed
Cancelled Services	Integer	Auto	LINQ Where + Count	ServiceStatusSummaryDto.Cancelled
Total Revenue	Decimal	Auto	LINQ Sum()	RevenueOverviewDto.TotalRevenue
Today Revenue	Decimal	Auto	LINQ Where + Sum	RevenueOverviewDto.TodayRevenue
This Month Revenue	Decimal	Auto	LINQ Where + Sum	RevenueOverviewDto.ThisMonthRevenue
Pending Payments	Decimal	Auto	LINQ Where + Sum	RevenueOverviewDto.PendingPayments
Total Customers	Integer	Auto	LINQ Where + Count	DashboardReportDto.TotalCustomers
Total Vehicles	Integer	Auto	LINQ Count()	DashboardReportDto.TotalVehicles
Total Technicians	Integer	Auto	LINQ Where + Count	DashboardReportDto.TotalTechnicians
Low Stock Parts	Integer	Auto	LINQ Where + Count	DashboardReportDto.LowStockPartsCount

17. Reports Screen

Field Name	Data Type	Mandatory	Validation Rules	Model/DTO
Year Filter	Integer	No	Optional year selection	Query Parameter
Month Filter	Integer	No	Optional month selection (1-12)	Query Parameter
Category Filter	Integer	No	Optional category ID	Query Parameter
Technician Filter	Integer	No	Optional technician ID	Query Parameter
Date Range From	DateTime	No	Start date for filtering	Query Parameter
Date Range To	DateTime	No	End date for filtering	Query Parameter

18. Notifications Screen

Field Name	Data Type	Mandatory	Validation Rules	Model/DTO
Notification ID	GUID	Auto	Primary Key, Auto generated	NotificationDto.Id
Title	String	Auto	Generated based on notification type	NotificationDto.Title
Message	String	Auto	Generated based on event	NotificationDto.Message
Type	String	Auto	Enum: ServiceStatusChange, Assignment, Payment, etc.	NotificationDto.Type
Is Read	Boolean	Auto	Default: false	NotificationDto.IsRead
Created At	DateTime	Auto	Auto generated on creation	NotificationDto.CreatedAt

Time Ago	String	Auto	Computed (e.g., "2 hours ago")	NotificationDto.TimeAgo
Related Entity ID	Integer	No	Link to related entity	NotificationDto.RelatedEntityId
Related Entity Type	String	No	Type of related entity	NotificationDto.RelatedEntityType
Action URL	String	No	Navigation link	NotificationDto.ActionUrl